

Apollo-RTOS: An AGC-Inspired Real-Time Operating System for Embedded Systems



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Abstract

Embedded systems have evolved dramatically, yet many of the challenges faced by early computing systems remain relevant today. The Apollo Guidance Computer (AGC), despite its limited hardware—a 2 MHz CPU, 4 KB of RAM, and 72 KB of ROM—introduced innovative scheduling and fault-tolerant mechanisms that ensured reliability in mission-critical environments. Inspired by the AGC, this project presents Apollo-RTOS, a real-time operating system (RTOS) for the Microbit v1, designed to explore the applicability of these historical concepts in modern embedded computing.

Apollo-RTOS implements a hybrid scheduling algorithm, combining cooperative and pre-emptive multitasking to optimize resource utilization and responsiveness. Additionally, it features a fault-tolerant recovery system, ensuring process integrity even during failures. The system extends the AGC's original design by incorporating modern OS functionalities, including a file system leveraging flash memory and an external FRAM device, inter-process communication through signals, and a shell interface for system interaction. Implemented in modern C++, Apollo-RTOS utilizes RAII, templates, and type deduction to enhance efficiency and maintainability.

This project demonstrates how historical embedded computing techniques can be adapted to contemporary microcontrollers, offering insights into designing resilient, efficient, and resource-aware real-time operating systems for modern applications.

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§1 Introduction

1.1 Background & Motivation

Computing capabilities have grown at an unprecedented rate over the last century. The **Apollo Guidance Computer** (AGC), which played a critical role in the Apollo program, operated with a modest *2 MHz CPU, 4 KB of RAM, and 72 KB of ROM*. To maximize its limited resources, the AGC's operating system introduced innovative solutions for *task scheduling, error handling, and resource-efficient computation*. One of its most groundbreaking features was a hybrid scheduling mechanism that combined cooperative and preemptive multiprogramming, ensuring that critical tasks could execute reliably even in adverse conditions. Additionally, the AGC incorporated a robust recovery mechanism, enabling fault tolerance in the face of system failures.

Today, a compact `micro:bit` device costing less than £20 possesses significantly more computing power than the AGC. Such microcontrollers are now ubiquitous, powering applications ranging from consumer electronics to mission-critical systems. Despite the vast improvements in computational power, many of the challenges that the AGC's operating system addressed remain highly relevant, particularly in constrained embedded environments where real-time performance, reliability, and fault tolerance are essential.

1.2 Project Objectives

The goal of this project is to study the design principles of the AGC's operating system and implement a robust real-time operating system (RTOS) for the `micro:bit v1`, named **Apollo-RTOS**. This project aims to explore the *applicability of the AGC's novel scheduling techniques and fault-tolerant mechanisms* in modern embedded computing. Additionally, it extends these foundational ideas by incorporating *contemporary operating system concepts* to enhance usability, efficiency, and resilience.

Apollo-RTOS integrates:

- **Hybrid scheduling** that balances cooperative and preemptive multitasking.
- **A fault-tolerant system** capable of handling hardware failures and ensuring process recovery.
- Separate **file systems** implemented on flash memory and an external FRAM device for persistent storage.
- **Inter-process communication** via signals for dynamic event handling.
- **A shell interface** for interacting with the system and managing processes.
- **A modern C++ implementation** leveraging RAII, templates, and type deduction for robustness and efficiency.

By bridging historical embedded computing principles with modern software engineering techniques, this project provides insights into designing efficient, fault-tolerant operating

systems for resource-constrained environments.

§2 The Apollo Guidance Computer

The Apollo Guidance Computer (AGC) was an early digital computer developed in the 1960s to help navigate and control Apollo spacecraft on their missions to the Moon. It provided real-time guidance and control, making it essential for the success of the Apollo program.

The AGC ran at a speed of 2.048 MHz, which translated to about 40,000 instructions per second. The AGC had a 16-bit word length, 15 bits for data, and one parity bit. It had 2 kilowords or 4KB of erasable magnetic-core memory (RAM) along with 36 kilowords or 72KB of fixed core rope memory (ROM) for storing software and mission data. Astronauts interacted with the AGC using the Display and Keyboard (DSKY), a 21-digit display with a 19-button keyboard that allowed them to input commands and receive information. This interface was critical for making manual adjustments and monitoring the system during missions.

Due to these constraints, the operating system, called the Executive, had to efficiently manage resources. It ensured that important tasks were prioritized over less critical ones and that key functions could continue even if the system experienced a reset due to an error.

The Executive was written in AGC assembly, where each instruction was 16 bits long. Because the instruction set was small, the system also included a sophisticated interpreter that acted as a virtual machine, providing more advanced pseudo-instructions beyond the basic AGC instructions.



Figure 0.1: Margaret Hamilton standing next to listings of the software she and her MIT team produced for the Apollo Project.

The details of the AGC Executive explained below is taken from O'Brien [2].

2.1 The Executive

The central component of the AGC software was the Executive, which ensured fair allocation of system resources, such as CPU time and dynamic memory, to processes and tasks. To seamlessly perform real-time control and navigation tasks with extreme reliability and efficiency, the Executive implemented a **priority-based cooperative scheduling** algorithm, described below:

- The highest-priority process has exclusive access to the CPU.
- Lower-priority processes remain idle until the highest-priority process:
 - lowers its priority, or
 - goes to `sleep` while waiting for an event.
- If a new process with a higher priority is scheduled while another is running:
 - the new process waits until the current process lowers its priority or goes to `sleep`,
 - then the new process starts executing.

Core Sets Information about running processes was stored in a table called Core Sets, which functioned similarly to modern process descriptors. Each Core Set had a structure similar to:

```

1 struct CORE_SET {
2     word priority;
3     word program_counter;
4     word bbank_reg; // area in unswitched memory where the code is
5     word vac_ptr;   // pointer to extra memory
6
7     word mpac[7];   // multi-purpose accumulators
8     word mpac_mode;
9 };

```

Each active process occupied one Core Set. The system provided:

- 6 Core Sets in the Command Module, and
- 7 Core Sets in the Lunar Module.

In either case, Core Set 0 always contained the currently running process. A context switch was essentially performed by swapping the current Core Set with Core Set 0.

Addressable Memory The 7 `mpac` words served two purposes. In normal mode, they provided 7 words of freely usable memory. In interpreter mode, they functioned as accumulator registers, storing values for calculations performed by the interpreter. If a process required more memory, it could request 42 additional words from the **Vector Accumulator (VAC)** area.

Starting a New Process To start a new process, an empty Core Set had to be located, followed by a VAC area if needed. After populating the Core Set, the Executive determined if the new process had a higher priority.

A special register, **NEWJOB**, was used for this purpose:

- If **NEWJOB** is 0, the current process has the highest priority.
- If **NEWJOB** is non-zero (say d), the process represented by Core Set d has the highest priority.

When a new process was created, its priority was compared to that of the currently running process, and **NEWJOB** was updated accordingly.

Process Switching A context switch only occurred when the current process voluntarily relinquished control. The process was required to check **NEWJOB** at least every 20ms to detect higher-priority processes. If **NEWJOB** was not checked for over 640ms, it was assumed that the process had stalled, triggering a hardware restart.

To switch processes, the Executive stored the current context in Core Set 0 and swapped it with the Core Set indexed by **NEWJOB**. Since the program counter was restored directly from the Core Set, execution of the switched-in process resumed automatically from its last instruction.

If the current process was still the highest-priority process but needed to go to sleep or lower its priority, the system scanned the Core Set table to identify the next highest-priority process and performed a switch accordingly.

Sleep and Wakeup A process could voluntarily relinquish execution while waiting for an external event by going to sleep. The **JOB_SLEEP** instruction marked a process as sleeping by negating its priority. The next highest-priority process was then scheduled.

When the awaited event occurred, another process could wake the sleeping process using the **JOB_WAKE** instruction. This reversed the priority negation, making the process runnable again. **JOB_WAKE** then checked **NEWJOB** to determine if the woken process should run immediately.

A process could also call **DELAYJOB** to sleep for a fixed duration. This scheduled an "alarm" task via the **waitlist**, described in the next section.

2.1.1 The Waitlist

The core processes in the Executive were all cooperative, meaning that even the highest-priority process would not preempt the currently running process until it voluntarily relinquished execution. The waitlist introduced preemptive scheduling capabilities to the AGC, allowing tasks to be scheduled with strict deadlines. Every 10ms, a timer interrupt was triggered to execute any waitlist task that was due.

In essence, the waitlist was a *sorted linked list* of tasks ordered by increasing deadlines. When a new waitlist task t was scheduled with a particular deadline, it was inserted into the list such that:

- any task t_0 with an earlier deadline appeared before t , and

- any task t_1 with a later deadline appeared after t .

Every 10ms, the timer interrupt checked whether the first task in the list had reached its deadline. If so, the task was executed and then removed from the list. Since waitlist tasks were executed within an interrupt handler, they had to be short-lived—typically around 5ms or 150–200 instructions.

The waitlist implementation in Apollo-RTOS closely follows that of the AGC. Implementation details will be discussed later.

2.2 Restart Mechanism

Despite its limited resources, the AGC needed to be highly reliable, even in the presence of errors. Modern CPU and OS designs dedicate significant resources to detecting and recovering from faults. However, the AGC lacked the hardware to support sophisticated error handling. Furthermore, without a memory protection unit, it was nearly impossible to reliably detect errors caused by memory corruption. As a result, the AGC adopted a radical approach:

Just restart everything.

However, restarting at an arbitrary point in time required careful handling, as it could terminate critical tasks mid-execution.

To address this, critical tasks and processes could define recovery behavior in case of a restart. Each program belonged to one of six restart groups. A process could use the `PHASCHNG` instruction to set up an entry in the restart phase table. Each group could recover only one process at a time. Based on its needs, a process could configure its group's recovery procedure to one of the following four modes:

- **Inhibit group restart:** No processes in this group are restarted.
- **Restart last display:** Restore and display the last known contents before the restart.
- **Execute one restart routine:** Restart the process from the beginning with minimal cleanup.
- **Execute two restart routines:** Schedule a task to clean up data and re-enter the process from a designated point.

To conserve memory, the AGC stored recovery information in a compressed format within the phase tables.

During a restart:

- All active jobs were canceled, and all tasks in the waitlist queue were deleted.
- Interrupts were disabled, and all tables were cleared and reinitialized.
- The integrity of the restart phase table was verified.
- Restart processing was carried out sequentially for each active phase in groups 6 to 1.

These mechanisms ensured that the AGC could gracefully recover from many types of fail-

ures—an essential requirement for spacecraft operations, where computer failures could have catastrophic consequences.

2.3 Interpreter

The Apollo Guidance Computer (AGC) featured a custom virtual machine known as the **Interpreter**, designed to extend its limited instruction set. This was necessary because the AGC's native instruction set was highly restricted, containing only 11 opcodes in AGC Block II. The Interpreter enabled more complex operations while conserving memory and improving execution efficiency for key tasks.

The Interpreter functioned as a *bytecode interpreter*, running on top of the AGC's native instruction set. It allowed the execution of *pseudo-instructions*, which were more powerful than basic machine instructions. These pseudo-instructions were stored as compact *6-bit opcodes*, maximizing memory efficiency.

The AGC Interpreter provided an extended set of *arithmetic and vector operations*, primarily aiding navigation and guidance computations. Key capabilities included:

- **Arithmetic Operations:** Fixed, single, and double-precision floating-point arithmetic, as well as `sqr` calculations.
- **Vector and Matrix Operations:** Dot products, cross products, and matrix transformations.
- **Trigonometric Functions:** Sine, cosine, and related functions for celestial navigation.
- **Data Movement and Manipulation:** Indexed memory access (not available in native AGC instructions), indirect addressing, and array operations.
- **Flow Control:** Conditional branching and subroutine-like operations for reusable code execution.

The AGC's native instruction set prioritized speed but lacked flexibility for advanced navigation tasks. The Interpreter enabled *higher-level mathematical operations* to be executed compactly and efficiently.

2.4 Display and Keyboard

The AGC featured a unique human-machine interface known as the *Display and Keyboard (DSKY)*, allowing astronauts to communicate with the computer using an intuitive *verb-noun paradigm*. With its green *electroluminescent displays*, *19-key input panel*, and *status indicators*, the DSKY played a crucial role in both the Command Module and Lunar Module cockpits. Despite its simple appearance, it incorporated sophisticated software routines, enabling astronauts to *monitor spacecraft systems*, *compute trajectories*, and *execute maneuvers with precision*.

The *verb-noun paradigm* structured commands similarly to natural language, where astronauts first selected a *verb* (the action) followed by a *noun* (the object of that action). Using

two-digit numeric codes for both components, astronauts could issue commands efficiently. For example, the command “*Verb 06 Noun 36*” instructed the AGC to *display the current time*. This system provided a consistent interface for a range of operations, from *simple data retrieval* to *complex trajectory calculations*.

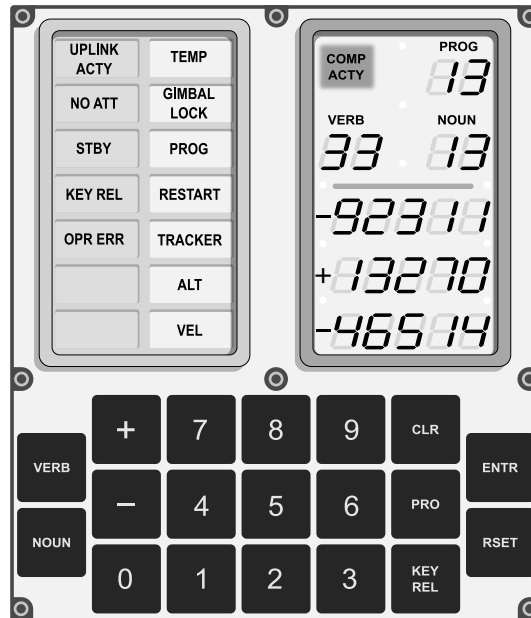


Figure 0.2: DSKY interface diagram by Oona Räisänen

§3 Inspiration behind Apollo-RTOS

The real-time operating system developed in this project, Apollo-RTOS, draws inspiration from both the Apollo Guidance Computer (AGC) operating system and the modern Linux OS. While the scheduler and error handling concepts are derived from the AGC, the naming conventions, user interface, and implementation are influenced by Linux.

The AGC OS was designed to meet specific mission requirements, allowing for a simplified and optimized design. In contrast, Linux is a general-purpose OS, designed to support applications with vastly different needs, making its architecture more complex.

The AGC operated within a well-defined hardware and software environment, where all components could be trusted. As a result, isolating the hardware, OS, and application code was unnecessary. Additionally, the AGC lacked memory protection features, making kernel/user space isolation, a fundamental aspect of general-purpose operating systems like Linux, impossible.

In Apollo-RTOS, we adopted the same high-level design principles as the AGC. Our goal was to implement its key innovations in a modern context while ensuring the OS could function efficiently in a resource-constrained environment. To achieve this, we made the following design choices:

- **Hardware Selection:** The `micro:bit v1` was chosen as the primary target platform. Its Cortex-M0 CPU is one of the smallest in the ARM family and closely resembles the AGC hardware. Additionally, it includes pre-installed peripheral devices, which serve as useful test cases for OS functionality.
- **No Kernel/User Space Isolation:** In microcontroller programming, the entire application is compiled into a single binary, meaning the kernel and user space are inherently combined. Since the entire codebase has full access to the kernel internals, enforcing strict isolation would add unnecessary complexity.
- **Monolithic Kernel Design:** Inspired by Linux, Apollo-RTOS follows a monolithic kernel architecture due to its simpler implementation and better execution efficiency in a constrained environment.
- **Library Calls Instead of System Calls:** Traditional system calls are beneficial only when user and kernel space are strictly separated. Since Apollo-RTOS does not enforce such isolation, we opted for direct library function calls, significantly simplifying implementation.
- **Shared Memory for Inter-Process Communication (IPC):** While shared memory introduces risks like race conditions and synchronization issues, it remains the fastest and simplest IPC method, making it the preferred choice for Apollo-RTOS.
- **Liberty to Use Memory Liberally:** While the `micro:bit` has limited memory, it still offers significantly more than the AGC. As Donald Knuth famously said, "premature optimization is the root of all evil," our priority is a functional and maintainable implementation rather than aggressively minimizing memory usage.

Additionally, we drew inspiration from the `microbian` OS for the `micro:bit`, as developed

by Spivey [5]. Specifically, we referenced its hardware description headers, startup code, and interrupt handler implementations in certain parts of our design.

For our language and development ecosystem, we chose to work with C++ for several reasons:

- C++ reduces boilerplate code compared to C with its concise syntax for object creation, templates, and RAII paradigm.
- The C++ compiler provides extensive optimizations through templates, `constexpr` variables and functions, and references.
- C++ templates and type deduction allow for cleaner, more expressive code, improving development experience.
- And finally, a personal preference: *I just wanted to learn more C++.*

§4 Implementation Architecture and Design

Apollo-RTOS is a real-time operating system inspired by the AGC, designed for the `micro:bit v1` device. It follows a monolithic kernel architecture and is built around a cooperative scheduler with preemptive capabilities.

This section provides an overview of the core modules in Apollo-RTOS and their interconnections, followed by a discussion of the codebase structure.

4.1 Overview

The major modules of Apollo-RTOS are:

Scheduler The core of the OS, responsible for:

- Managing process execution, including creation, termination, priority changes, pausing, and resuming.
- Handling context switching between processes.

Memory Manager Handles all dynamic memory allocation. It implements a freelist-based `malloc` for allocating and deallocating runtime memory and manages process stack allocations.

Recovery Ensures system stability by protecting critical processes and tasks from various resets and failures.

Waitlist Implements a deadline-driven task scheduler, similar to cron jobs, enabling time-sensitive operations to be executed at precise intervals.

Drivers Interface with the hardware components of the `micro:bit`. The key drivers include:

- **Timer:** Apollo-RTOS utilizes two 16-bit timers—`Timer1` as a system clock and `Timer2` for profiling.
- **Serial:** Manages UART communication.
- **I2C:** Handles communication with the I2C master on the `micro:bit`.
- **Display:** Controls the 5×5 LED matrix on the board.

File System Provides persistent storage, implemented using an external FRAM over I2C. An optional Flash-based file system can also be enabled.

Shell Implements a Unix-like command-line interface that allows users to manage and debug processes over UART. It maintains a list of available commands mapped to various processes.

Signals and Mutexes Enable low-overhead inter-process communication (IPC) and synchronization mechanisms among the processes.

Debugging and Error Handling Provides essential debugging utilities, including:

- **debug**: Outputs diagnostic information.
- **assert**: Handles runtime error detection.

4.2 Codebase Structure

The codebase follows a standard structure, with headers residing in the `include/` directory and implementations in `src/`. These directories are further divided into subdirectories to group related files.

A brief overview of the key directories and files is as follows:

- **include/**: Contains headers for core functionality.
 - **core/**: Core system components.
 - `hardware.h` — Hardware register descriptions (adapted from Spivey [5]).
 - `types.h` — Type aliases and constants shared across the codebase.
 - `irq.h` — Header-only interrupt handling utilities.
 - `boot.h` — Functions for retrieving the current boot level.
 - `sched.h` — Scheduler interface.
 - `waitlist.h` — Methods for registering waitlist tasks.
 - `memory.h` — Memory management utilities.
 - `recover.h` — Recovery mechanisms for processes and tasks.
 - `signal.h` — Methods for inter-process signaling.
 - **drivers/**: Hardware drivers.
 - `timer.h` — Provides system time (now) and profiling (`debug_ticks`).
 - `display.h` — Convenience functions for displaying images on the LED matrix.
 - `i2c.h` — Methods for I2C communication.
 - `accel.h` — Reads accelerometer data.
 - `gpio.h` — GPIO utility functions.
 - `serial.h` — Serial I/O functions, including an interface for redirecting input from the shell to other processes.
 - **utility/**: General-purpose utilities.
 - `allocator.h` — Core memory pool allocator used by different `malloc` implementations.
 - `format.h` — Provides printf-style formatting functions.

- `debug.h` — Conditional debugging output based on verbosity levels; also includes the `assert` macro.
- `mutex.h` — Lightweight mutex implementation using wait queues and `sleep/wakeup` from `sched.h`.
- `profile.h` — Macros for enabling function and process profiling.
- `args.h` — Data structures for passing command-line arguments to processes.
- `bitset.h` — Bitset implementation.
- `circular_buffer.h` — Circular buffer implementation.
- **fs/**: Header-only file system implementation (detailed later).
 - `fs_bck.h` — Backend for reading/writing device storage.
 - `fs_impl.h` — Manages open files, runtime memory maps, and file access.
 - `fs_desc.h` — Type definitions for template-based file system operations.
 - `fs.h` — Top-level file system interface, enabling separate instances for FLASH and FRAM storage.
- **src/**: Mirrors `include/`, implementing the corresponding functionality.
 - `vector_table.c` — Defines the interrupt vector table.
 - `mpx.s` — Handles context switching logic.
 - `device.ld` — Linker script defining memory layout.
 - `startup.cpp` — Initializes data sections and sets up core system modules.
- **apollo/**: Defines processes and built-in shell commands for Apollo-RTOS.
- **tests/**: Contains test cases demonstrating various aspects of the system.

Language	Files	Lines	Code	Comments	Blanks
Assembly	1	98	55	28	15
C	1	114	83	11	20
C++ Header	31	3402	2646	120	636
C++	27	2694	2128	75	491
Total	60	6308	4912	234	1162

§5 Scheduling: Co-operative and Pre-emptive

The core scheduler of Apollo-RTOS is a priority-based cooperative scheduler. It allows the highest-priority process to run indefinitely until it voluntarily gives up execution, at which point other processes are given the opportunity to run. This simple approach offers several benefits:

- **Simplicity:** The implementation is straightforward.
- **Fewer race conditions:** Since one process never preempts another, data races on shared variables are largely eliminated.

However, processes must be aware of their runtime context and yield execution periodically to prevent deadlocks. This can be done in two ways:

- Lowering their priority.
- Going to sleep.

With this design, the interface for process creation and management remains simple:

```

1 // Register a process defined by 'proc_def', passing 'param' as an
   argument
2 pid_t reg_proc(const proc_def_t *proc_def, void *param);
3
4 // Increase the priority
5 void incr_priority(pid_t pid, priority_t priority);
6
7 // Decrease the priority
8 void decr_priority(priority_t priority);

```

Defining a process is minimalistic:

```

1 struct proc_def_t {
2     string      name;
3     priority_t  priority;
4     size_t      stk_sz;
5     runnable_t  func;          // where runnable_t = void ()(void);
6 };

```

At runtime, each process is represented in a process descriptor table, which stores essential metadata about active processes. The most important fields are:

```

1 struct proc_t {
2     priority_t  priority;      // Current priority
3     byte_t      *stk_ptr;      // Current stack pointer
4     byte_t      *stack;        // Start of the stack area
5     size_t      stk_sz;        // Stack size
6     void        *param;        // Parameter passed to the process function
7     // ... other fields
8 };

```

Stack Layout in Arm Cortex-M0

To understand the scheduling mechanism, we first examine the stack layout in the Arm Cortex-M0 architecture:

The stack in Arm architecture starts at a higher address and grows downward.

When execution is preempted by an interrupt, the hardware automatically pushes key registers onto the stack in the following order (from lower to higher addresses):

```

new stack pointer -> |
                    | r0
                    | r1
                    | r2
                    | r3
stack grows      ^   |
downwards       |   | r12
                |   | lr      Link register
                |   | pc      Program counter
old stack pointer -> |   | psr    Status register

```

To enable context switching between processes, additional registers must be saved manually. This is done via software, giving us control over the order. We chose the following layout:

```

new stack pointer -> |
                    | r8
                    | r9
                    | r10
                    | r11
                    | r4
                    | r5
                    | r6
stack pointer after | r7
hardware pushes regs -> | lr_intr    Return from interrupt address

```

With the stack management details covered, we can now discuss how the scheduling algorithm works.

5.1 Process Creation

The implementation of **reg_proc** can be summarized as follows:

```

src/core/sched.cpp
1 pid_t reg_proc(const proc_def_t *def, void *param)
2 {
3     auto [name, priority, stk_sz, proc_func] = *def;
4     proc_t *proc = procs.alloc(); // allocate a process descriptor
5     proc->param = param; // store the param for future cleanup
6
7     stack_allocator.acquire(proc, stk_sz, proc_func, param, exit);
8     incr_priority(pid, priority);
9
10    // ...
11    return pid;

```

12 }

The key operations occur on lines 7 and 8. In line 7, the local pool allocator assigns a block of memory of size `stk_sz` and initializes it with the required parameters.

A newly created process is treated as if it has just undergone a context switch. This means the stack must be set up with the appropriate register values in the correct order, as discussed earlier. Line 7 ensures this setup is performed correctly.

Examining the `acquire` function:

```

src/core/stack_allocator.h
1 void acquire(proc_t *proc, size_t stk_sz, runnable_t func, void *param,
2             void (*ret)())
3 {
4     proc->stack = (byte_t *)pool.alloc(stk_sz); // allocate the stack
5     // ...
6
7     auto stk_ptr = (context_t *) (proc->stack + stk_sz - sizeof(context_t));
8
9     // setup initial stack frame
10    stk_ptr->psr    = init_psr;           // 0x01000000 to enable Thumb mode
11    stk_ptr->pc     = (uint32_t)func;      // entry to the process
12    stk_ptr->lr     = (uint32_t)ret;       // exit from the process
13    stk_ptr->r0     = (uint32_t)param;     // parameter
14    stk_ptr->lr_intr = lr_intr_magic;      // return address from interrupts
15
16    proc->stk_ptr = (byte_t *)stk_ptr;
17    proc->stk_sz = stk_sz;
18 }

```

Lines 10-14 populate the process context. The entry point is stored in the pc register, the exit sequence in the lr register, and the process parameter in r0. The interrupt return address is also set.

5.2 Priority Modification

A singleton object, `cpu`, tracks the currently running process and the highest-priority process:

```

src/core/sched.cpp
1 volatile struct {
2     proc_t *hi_proc = nullptr; // highest priority process
3     proc_t *curr_proc = nullptr; // currently running process
4 } cpu;

```

This object functions similarly to the `NEWJOB` register in the AGC. When a process is created, its priority is compared with the highest-priority process. If it is higher, `hi_proc` is updated.

In `reg_proc` (line 8), `incr_priority` is called:

```

src/core/sched.cpp
1 void incr_priority(pid_t pid, priority_t priority)
2 {
3     proc_t *proc = procs[pid];
4     // ... disable interrupts
5     proc->priority = priority; // set priority
6     if (cpu.hi_proc == nullptr || cpu.hi_proc->priority < priority)
7         cpu.hi_proc = proc;
8 }

```

When a high-priority process completes its critical section, it should lower its priority to allow other processes to run:

```

src/core/sched.cpp
1 void decr_priority(priority_t priority)
2 {
3     cpu.curr_proc->priority = priority;
4     cpu.hi_proc = procs.max_priority();
5     change_proc();
6 }

```

The highest-priority process is updated, and **change_proc** is called to check if a context switch is needed:

```

src/core/sched.cpp
1 void change_proc()
2 {
3     // enable interrupts
4     if (cpu.hi_proc != cpu.curr_proc)
5         reschedule(); // triggers pendsv interrupt
6 }

```

Now, we examine how context switching is performed.

5.3 Context Switching

When `reschedule()` triggers the `pendsv` interrupt, the hardware first pushes key registers onto the stack. Then, `pendsv_handler` is called. We then save additional registers, swap stack pointers, and restore the state for the new process.

The register saving/restoring is handled in assembly, while stack pointer swapping is done in C++:

```

src/mpx.s
1 @@@ push software-saved registers onto stack
2 .macro isave
3     push {r4-r7, lr}           @ Copy from high to low
4     mov r4, r8
5     mov r5, r9
6     mov r6, r10
7     mov r7, r11

```

```

8  push {r4-r7}
9  mrs r0, msp                @ Copy from high to low
10 .endm
11
12 @@@ pop software-saved registers
13 .macro irestore              @ Expect process sp in r0
14 msr msp, r0
15 pop {r4-r7}
16 mov r11, r7
17 mov r10, r6
18 mov r9, r5
19 mov r8, r4
20 pop {r4-r7}
21 pop {pc}                    @ pop lr into pc, jumping to new process
22 .endm
23
24 @@@ PendSV handler for context switching
25 .global pendsv_handler
26 .thumb_func
27 pendsv_handler:
28     isave                    @ Save current process state
29     bl cxt_switch            @ Select new process
30     irestore                 @ Restore new process state

```

The definition of **cxt_switch**:

```

src/core/sched.cpp
1 void *cxt_switch(void stk_ptr)
2 {
3     if (/* cpu.curr_proc is done */)
4         /* release resources */
5     else
6         cpu.curr_proc->stk_ptr = (byte_t *)stk_ptr; // store old stk_ptr
7
8         cpu.curr_proc = cpu.hi_proc;
9         return cpu.hi_proc->stk_ptr; // restore new stk_ptr
10 }

```

5.4 Process Suspension

A process can voluntarily yield execution either by lowering its priority or by suspending execution (sleeping) to be woken up later.

The wake-up mechanism is more involved, so we will discuss it in two parts. A process can wake up in one of two ways:

- By setting an alarm when going to sleep, which wakes it up after a specified time.
- By waiting for a specific event to occur before resuming execution.

The first method relies on the waitlist, while the second requires a synchronization mechanism. We will first discuss process suspension before covering these mechanisms.

A runnable process is represented by a positive priority value. To put a process to sleep while preserving its original priority, we simply negate its priority.

src/core/sched.cpp

```

1 void sleep(time_t period = 0)
2 {
3     if (period > 0)
4         /* set an alarm in the waitlist */
5         /* else suspend execution until another process wakes me up */
6
7     // Negate priority to mark process as ASLEEP
8     decr_priority(-cpu.curr_proc->priority);
9 }

```

A process that wants to wake up a sleeping process simply negates its priority:

src/core/sched.cpp

```

1 void wakeup(pid_t pid) {
2     incr_priority(pid, -procs[pid]->priority);
3 }

```

5.5 Waitlist: Preemptive Scheduling

The waitlist is implemented as a sorted linked list of tasks:

src/core/waitlist.cpp

```

1 struct waitlist_t {
2     time_t remaining = 0;
3     runnable_t func = nullptr;
4     void *param = nullptr;
5     waitlist_t *next = nullptr; // Linked list link
6 };

```

If a task t_0 is scheduled to run in n_0 ticks, it is placed in the waitlist between tasks t_1 and t_2 such that if t_1 and t_2 have deadlines n_1 and n_2 , then $n_1 \leq n_0 \leq n_2$. We set:

$$\text{remaining} = n_0 - n_1$$

Thus, the time until executes is the sum of the remaining values for all preceding tasks.

This algorithm, originally used in the AGC, has key implications:

- Maintaining a sorted linked list has insertion complexity. However, with an upper limit of $N_WAITLIST = 16$, this cost is negligible.
- Using remaining values, we only need to check the first task on each timer tick, reducing computational overhead.

Task registration and execution are implemented as follows:

src/core/waitlist.cpp

```

1 void reg(string name, time_t interval, runnable_t func, void *param)
2 {
3     waitlist_t *task = alloc();
4     waitlist_t *hd = &waitlist_hd;
5     time_t prev = 0;
6     while (hd->next != nullptr && prev + hd->next->remaining <= interval) {
7         prev += hd->next->remaining;
8         hd = hd->next;
9     }
10    time_t remaining = interval - prev;
11
12    if (hd->next != nullptr)
13        hd->next->remaining -= remaining;
14
15    *task = {name, remaining, func, param, hd->next};
16    hd->next = task;
17 }

```

During a timer interrupt, expired tasks are executed:

```

src/core/waitlist.cpp
1 void run()
2 {
3     auto head = &waitlist_hd;
4     if (head->next != nullptr)
5         head->next->remaining -= update_interval;
6
7     while (head->next != nullptr && head->next->remaining == 0) {
8         auto task = head->next;
9         head->next = task->next;
10
11         auto [name, remaining, func, param, next] = *task;
12         dealloc(task);
13         func(param);
14     }
15 }

```

This approach minimizes overhead in interrupt-inhibited situations, making it ideal for real-time constraints.

5.5.1 Wakeup Alarm

With the waitlist implemented, setting up an alarm is simple:

```

src/core/sched.cpp
1 void sleep(time_t period = 0)
2 {
3     auto proc = cpu.curr_proc;
4
5     auto default_alarm = [](void *ptr) {
6         auto proc = (proc_t)ptr;

```

```

7   incr_priority(/ pid of proc */, -proc->priority);
8   };
9
10  if (period > 0)
11    waitlist::reg(proc->name, period, default_alarm, (void *)proc);
12
13  decr_priority(-proc->priority);
14 }

```

5.6 Synchronization Mechanism

We implement channels for synchronization, using a FIFO queue of waiting processes based on `circular_buffer`:

```

                                include/core/sched.h
1  template <size_t chan_len>
2  struct chan_t : public circular_buffer<pid_t, chan_len> {};

```

A process waiting for an event enqueues itself and sleeps, while another process signals the event by dequeuing and waking it up:

```

                                include/core/sched.h
1  template <size_t chan_len>
2  void wait(chan_t<chan_len> &chan)
3  {
4    intr_guard guard;
5    chan.enqueue(/* current process */);
6    sleep();
7  }
8
9  template <size_t chan_len>
10 void notify(chan_t<chan_len> &chan)
11 {
12   intr_guard guard;
13   if (chan.empty())
14     return;
15   wakeup(chan.dequeue());
16 }
17
18 template <size_t chan_len>
19 void notify_all(chan_t<chan_len> &chan)
20 {
21   intr_guard guard;
22   while (!chan.empty())
23     wakeup(chan.dequeue());
24 }

```

Using channels and `wait`, `notify`, we can implement a `mutex`

```

                                include/utility/mutex.h
1  template <size_t chan_len>

```

```

2 class mutex {
3     chan_t<chan_len> lock_chan;
4     bool busy = false;
5
6 public:
7     void lock()
8     {
9         while (busy)
10            sched::wait(lock_chan);
11        busy = true;
12    }
13
14    void unlock()
15    {
16        busy = false;
17        sched::notify(lock_chan);
18    }
19 };

```

Usually we acquire a mutex before a critical section, and then release it at the end of the section. Using the RAII paradigm, we can implement a class `lock_guard` (named after `std::lock_guard`) that acquire the mutex upon construction, and releases it in its destructor, giving a simple interface for locking.

```

                                include/utility/mutex.h
1 template <typename _mutex>
2 class lock_guard {
3     _mutex &_m;
4
5 public:
6     lock_guard(_mutex &m) : _m{m} { _m.lock(); }
7     ~lock_guard() { _m.unlock(); }
8 };

```

5.6.1 Interrupts management

We also implement a similar mechanism for disabling and enabling interrupts before and after a critical section. This implementation uses template type deduction to choose between the two scenarios when we want to disable all interrupts vs disable only one interrupt:

```

                                include/core/irq.h
1 template <typename T>
2 class intr_guard
3 {
4     int irq;
5
6 public:
7     // when no argument is passed, deduce T = void & disable all interrupts
8     intr_guard(void) requires std::same_as<T, void>
9         : irq{get_primask()}
10    {

```

```

11     intr_disable();                                |  T == void  |
12 }                                                  | all irqs    |
13                                                  +-----+
14 // re-enable all interrupts if T = void
15 ~intr_guard() requires std::same_as<T, void>
16 {
17     set_primask(irq);
18 }
19
20
21 // if an arugment is passed, deduce T = int, and disable only that irq
22 intr_guard(int irq) requires std::same_as<T, int>
23     : irq{irq}
24 {
25     disable_irq(irq);                                |  T == int   |
26 }                                                  | one irq     |
27                                                  +-----+
28
29 // if T = int, enable only that irq
30 ~intr_guard() requires std::same_as<T, int>
31 {
32     clear_pending(irq);
33     enable_irq(irq);
34 }
35 };
36
37 // type deduction hints for the constructor
38 intr_guard(void) -> intr_guard<void>;
39 intr_guard(int) -> intr_guard<int>;

```

This interface is then used simply as:

```

1 { // critical section
2     intr_guard guard; // disable all interrupts and re-enable at the end
3 }
4
5 { // critical section
6     intr_guard guard{UART_IRQ}; // disable only the UART interrupt
7 }

```

§6 Memory Management and File System

A central part of Apollo-RTOS is its memory allocation and management system. All run-time allocation is handled through instances of a pool allocator, which provides a simple and efficient form of dynamic memory management.

- The allocator maintains a linked list of free memory chunks.
- On an allocation request, it searches the free list for a chunk of sufficient size.
- If a suitable chunk is found, it is returned; otherwise, more memory is requested from the global allocator.
- When **free** is called, the chunk is returned to the free list.

```

                                include/utility/allocator.h
1  template <byte_t *(*alloc_func)(size_t), size_t alignment>
2  class allocator
3  {
4      chunk_t free_hd;
5      static constexpr size_t header_sz = sizeof(chunk_t);
6
7  public:
8      // allocate chunk of sz bytes
9      byte_t *alloc(size_t sz)
10     {
11         sz = roundup(sz, alignment);
12         auto chnk_sz = header_sz + sz;
13
14         chunk_t *chunk = /* find a chunk in free_hd, nullptr otherwise */;
15         if (chunk == nullptr) {
16             chunk = (chunk_t *)alloc_func(chnk_sz); // get new chunk
17             chunk->sz = sz;
18         } else {
19             /* remove chunk from free_hd */
20         }
21
22         return (byte_t *)chunk + header_sz;
23     }
24
25     // return chunk allocated by this allocator
26     void dealloc(byte_t *ptr)
27     {
28         if (ptr == nullptr)
29             return;
30         auto chunk = (chunk_t *) (ptr - header_sz);
31         /* insert chunk to free_hd */
32     }
33 };

```

Having an instantiable allocator instead of a global **malloc** function has key benefits:

- Since the hardware lacks a Memory Protection Unit (MPU), this helps isolate unrelated

memory allocations in software, reducing memory corruption risks.

- It provides a convenient interface for creating local pool allocators.

Using this allocator, we implement kernel-space memory management via **kmalloc** and **kfree**.

```
src/core/memory.cpp
1 namespace kmem
2 {
3     allocator<alloc_heap, 8> kpool;
4
5     void *kmalloc(size_t sz) { return kpool.alloc(sz); }
6
7     void kfree(void *ptr)
8     {
9         if (ptr == nullptr)
10             return;
11         kpool.dealloc((byte_t *)ptr);
12     }
13 } // namespace kmem
```

For user-space memory allocation, the implementation is more involved. When a process allocates memory, it is expected to free it explicitly. However, if the process encounters an error or is terminated prematurely, the scheduler must clean up any unfreed memory.

Each process descriptor maintains a list of allocated memory chunks. When a process terminates, all remaining allocations in this list are freed automatically.

```
src/core/memory.cpp
1 namespace heap
2 {
3     allocator<alloc_heap, 8> pool;
4
5     void *malloc(size_t sz)
6     {
7         byte_t *ptr = pool.alloc(sz);
8         /* insert chunk to the current process' allocated list */
9         return ptr;
10    }
11
12    void free(void *ptr)
13    {
14        if (ptr == nullptr)
15            return;
16        /* remove chunk from process' allocated list */
17        pool.dealloc((byte_t *)ptr);
18    }
19
20    // called at process teardown
21    void cleanup(chunk_t *hd)
22    {
23        while (hd->next != nullptr) {
```

```

24     auto ptr = (byte_t *)hd->next + sizeof(chunk_t);
25     /* remove hd->next from the list */
26     pool.dealloc(ptr);
27 }
28 }
29 } // namespace heap

```

6.1 File System

Flash The `micro:bit` features approximately 256KB of flash memory. A build of Apollo-RTOS with the `-O2` flag produces a binary of about 92KB, while using `-Os` reduces it to 76KB. In either case, there is ample remaining space in flash that can be used for long-term storage. However, flash memory has significant drawbacks:

- **Inefficient writes:** Writing to flash requires first erasing entire 1KB pages, taking about **22.3ms** per page. Writing individual bytes afterward takes **20μs** each [3].
- **Limited endurance:** The `micro:bit`'s flash memory has an endurance of only **20,000 erase cycles** and retains data for approximately **10 years** under normal conditions. Excessive writes degrade reliability.

Despite these limitations, reading from flash is fast since the CPU has direct access to it—after all, the program itself runs from flash. This makes flash memory suitable for read-heavy, infrequent-write storage needs.

FRAM Ferroelectric RAM (FRAM) addresses flash memory's shortcomings. Unlike flash, FRAM is non-volatile, with significantly higher endurance and retention rates. The MB85RC2567 FRAM chip [1] used in Apollo-RTOS provides:

- **10^{12} erase cycles** – far exceeding flash longevity.
- **95+ years** of data retention under normal conditions.
- **32KB capacity**, operating at 1MHz I2C, though we use 100KHz.

However, FRAM has its own trade-offs:

- **I2C communication overhead:** Since the `micro:bit` lacks integrated FRAM, it is connected as an external I2C peripheral. This introduces bus contention, affecting read/write performance.
- **Slower reads than flash:** Unlike flash, which the CPU can access directly, FRAM requires I2C transactions, making reads slower. However, writes are significantly faster than flash.

Since both storage options have their strengths and weaknesses, Apollo-RTOS implements file systems on both. Instead of maintaining two separate file systems, we designed an abstract block device that can be instantiated with different parameters for each storage medium.

A block device is characterized by:

- Constants defining block size, total blocks, etc.

- Read/Write methods for interacting with the device.

Since these constants are frequently used in file system operations, we implemented them as template parameters rather than function parameters. This allows the compiler to optimize operations more effectively.

Thus, our file system implementation is entirely header-only, residing in the `include/fs/` directory.

6.2 File system implementation

The file system consists of two layers of tables with extensive bookkeeping. Below, we outline key concepts and methods.

fs_desc_t defines the file system parameters. Two instances—one for flash, another for FRAM—are constructed at compile time and passed as template parameters to the implementation.

```

include/fs/fs_desc.h
1 template <...>
2 struct fs_desc_t {
3     using addr_t = /* type of address supported by device */;
4     // for flash, it is uint32_t since writes must be word aligned
5     // for fram, it can be uint16_t to save space
6
7     size_t blk_sz; /* size of a block */
8     // for flash, it is 1024,
9     // since that's the smallest unit we can effectively write
10    // for FRAM, it is 128
11
12    size_t max_num_blks; /* max number of blocks per file */
13    size_t n_inodes; /* number of files */
14
15    size_t n_open_files; /* number of open files */
16
17    // methods to transfer data to and from the target device
18    void (*write)(addr_t addr, uint8_t *buf, size_t buf_sz);
19    void (*read)(addr_t addr, uint8_t *buf, size_t buf_sz);
20 };

```

6.2.1 File system backend

Each file is represented by an **inode_t**, inspired by Linux inodes, containing file metadata.

```

include/fs/fs_bck.h
1 template <typename desc_t, desc_t desc>
2 struct inode_t {
3     using addr_t = typename desc_t::addr_t;
4     using blk_addr_t = uint8_t;

```

```

5     // in either flash and fram, we can refer to a block by a 8 bit uint
6
7     size_t nblks; /* number of blocks in this file */
8     blk_addr_t blks[desc.max_num_blks]; /* blocks in this file */
9
10    uint32_t flag; /* stats encoded into a single uint32_t */
11 };

```

We use `fn_t = uint8_t` as the file name. We currently don't map file name strings to the corresponding inode, and directly refer the file by the inode number.

The file system header maintains:

- A table of inodes.
- A **bitset** tracking free and occupied blocks.

```

                                include/fs/fs_bck.h


---


1 template <typename desc_t, desc_t desc>
2 struct fs_hd_t {
3     inode_t inode_tbl[desc.n_inodes];
4     bitset<N_BLKs> free_set; /* N_BLKs is total number of available blks */
5
6     bool valid() const;
7     void format(); // format the file system
8
9     // find the inode for the file fn
10    inode_t &find(fn_t fn);
11
12    // find nblks unclaimed blks and assign them to the given buffer
13    void alloc_blks(blk_addr_t *buf, size_t nblks);
14
15    // reclaim the given blks
16    void dealloc_blks(blk_addr_t *buf, size_t nblks);
17 };

```

The file system backend is then a wrapper around the header, that provides methods for:

- **open** and **remove** files.
- **load** and **store** data.
- **mount** and **format** operations.

```

                                include/fs/fs_bck.h


---


1 template <typename desc_t, desc_t desc>
2 class fs_t
3 {
4     fs_hd_t fs_hd;
5
6 public:
7     void format();
8     void mount(); /* if the header is not valid, it also formats it */
9     void umount() const;
10

```

```

11 // Open a file using Linux like flags O_CREATE, O_WRITE, O_SHARED etc.
12 const inode_t *open(fn_t fn, size_t sz, uint32_t flags);
13
14 void remove(fn_t fn);
15
16 // Load the contents of the file to the given buffer
17 void load(const inode_t *inode, uint8_t *buf, size_t buf_sz) const;
18
19 // Store the contents of the given buffer to the file
20 void store(const inode_t *inode, uint8_t *buf, size_t buf_sz) const;
21
22 // Store the contents of the buffer at a particular offset.
23 void store(const inode_t *inode, size_t off, uint8_t *buf,
24           size_t buf_sz) const;
25 };

```

load and **store** map buffer contents to blocks, allowing files to have non-contiguous block allocations.

Currently a file's size cannot be modified. But it should be fairly simple to increase the size of a file, just by allocating more blocks to it.

6.2.2 File System Frontend

The frontend maintains a table of open files and maps them to runtime memory addresses. This mapping can be either private or shared:

- Private mapping: A file is mapped to a runtime buffer local to the process.
- Shared mapping: Multiple processes access the same runtime representation of the file, enabling inter-process communication (IPC).

This mechanism is inspired by the `mmap` function in Linux.

File operations occur in the following stages:

Opening a File When a process requests to open a file `fn`:

- The frontend retrieves the associated inode, creating one if it doesn't exist.
- It locates an available file descriptor in the open files table.
- A file handler is wrapped around the descriptor and returned to the process.

Mapping a File The mapping strategy depends on whether the `O_SHARED` flag is set during the `open` call:

- Shared mapping:
 - A buffer is allocated, and its address is stored in the file descriptor.
 - If a mapping already exists, it is reused.
 - All **load** and **store** operations use this shared buffer.

■ Private mapping:

- A buffer is allocated and stored in the file handler.
- Other processes cannot access this mapping.

Loading and Storing a File Once mapping is established, all **load** and **store** operations reference the appropriate buffer from either the file descriptor (shared mapping) or file handler (private mapping).

The file descriptor maintains:

- A pointer to the underlying inode.
- A reference count for read and write operations.
- Information on shared memory mappings.

include/fs/fs_impl.h

```

1 class fd_t
2 {
3     uint8_t w_cnt, r_cnt; /* read and write reference counter */
4
5 public:
6     fn_t fn;
7     mutable mutex<8> mtx = {"fd"}; /* file operations should be atomic */
8     const inode_t *inode; /* the underlying file */
9     mmap_unit_t mu; /* mapping information */
10
11 public:
12     // if this file descriptor refers the file fn, increse ref count
13     // otherwise mark this descriptor in use
14     void acquire(fn_t _fn, const inode_t *_inode, bool w_en);
15
16     // decrease ref count,
17     // if it was the last reference to the file, close this descriptor
18     void release(bool w_en);
19 };

```

The file handler is what the process interacts with. It holds a pointer to the file descriptor and provides methods for interacting with the file, ensuring smooth access and management.

include/fs/fs_impl.h

```

1 class file_t
2 {
3     fd_t *fd;
4     mmap_unit_t mu; /* private mapping information */
5     const bool is_shared; /* is the mapping shared? */
6     const bool w_en = false; /* have write permission? */
7
8 public:
9     // open the file in an RAII style, close it in the desctructor
10    file_t(fn_t fn, size_t _sz, uint32_t flags);
11
12    ~file_t();

```

```

13  void close();
14
15  // create runtime mapping
16  void *mmap(size_t sz);
17
18  // instead of allocating new memory, use the given memory address
19  void mmap(uint8_t *buf, size_t sz);
20
21  // remove the mapping
22  void unmap();
23
24  void load();
25  void store();
26  // store a chunk of size sz at offset off
27  void store(size_t off, size_t sz);
28 };

```

Finally, the top-level file system implementation maintains:

- An array of open files.
- An `fs_t` instance referring to the backend.
- Methods for interacting with the file system backend.

```

                                include/fs/fs_impl.h
1  template <typename desc_t, desc_t desc>
2  class fs_impl_t
3  {
4      // the backend implementation
5      inline static fs_t fs;
6
7      // pool allocator providing memory for shared mapping
8      inline static allocator<alloc_heap, 4> pool;
9
10     // table of open files
11     inline static fd_t open_files[desc.n_open_files] = {};
12
13 public:
14     static file_t open(fn_t fn, size_t sz, uint32_t flags);
15     static void remove(fn_t fn);
16     static void mount();
17     static void format();
18 };

```

6.2.3 File system instantiation

So far, the file system has been an abstract concept implemented through a complex network of templates. To create actual file systems, we need to instantiate these templates with specific parameters. After defining the required parameters for both the flash and FRAM file systems, the instantiation process becomes straightforward:

```
include/fs/fs.h
```

```
1 // instantiate with the parameters for the flash fs
2 class flash : public fs_impl_t<_flash::desc_t, _flash::desc> { };
3
4 // the same for the fram fs
5 class fram : public fs_impl_t<_fram::desc_t, _fram::desc> { };
```

Now, we can use files as follows:

```
                                apollo/test.cpp
1 void file_rw_test()
2 {
3     fn_t fn = 10;
4
5     auto file = fram::open(fn, 16, O_WRITE | O_CREATE);
6     auto t = (char *)file.mmap(16);
7
8     const char *s = "It IS a string";
9     while (*s != '\0')
10         *t++ = *s++;
11     *t = '\0';
12
13     file.store();
14     // the file is automatically closed by the destructor
15 }
```

§7 Error Handling and Recovery

For a critical mission like the Apollo program, where astronauts' lives depended on flawless execution, thorough testing of the entire codebase was essential. However, no matter how well-tested a system is, some errors only show up randomly due to hardware faults or unforeseen corner cases. To handle such situations, the AGC's operating system had a robust error handling and recovery mechanism.

Because of the thorough testing during development, the AGC developers assumed that any error occurring at runtime was likely from some external source rather than a software bug. This made the age-old solution of "just rebooting it" a reasonable recovery strategy.

Of course, an unexpected restart during a critical operation could cause problems. To mitigate this, the AGC maintained a restart table, where processes could store recovery information, ensuring they could resume correctly after a reset.

Apollo-RTOS follows a similar strategy for error handling and recovery. Before explaining the implementation, we need to discuss how the system determines its current state.

7.1 Boot Levels

From an error-handling perspective, the device can be in one of four states:

- **FLASH** – The device has just been flashed. Everything is shiny and new, and all hardware has been reset.
- **POWER** – The device was unplugged and powered back on, resulting in a state similar to **FLASH**.
- **BOOT** – The reset button was pressed, causing a full restart, though some registers may retain their values.
- **RESET** – A software reset occurred. RAM contents are retained, but execution restarts from the beginning.

According to the Nordic nRF51 technical reference manual [4], RAM contents are retained in a soft reset, but may be corrupted in other forms of reset.

Now, suppose an error occurs. There are three ways to handle it:

- Terminate the faulty process and let the system continue.
- Trigger a soft reset, rebooting the entire system automatically.
- Halt and wait for manual intervention, requiring a physical reset.

A power outage is a more extreme case. In theory, once the device loses power, the mission has already failed. But since Apollo-RTOS isn't responsible for landing a spacecraft, we take a more lenient approach: when power is restored, the system attempts to resume major tasks that were running before the outage.

7.1.1 How to Detect Boot Stages

To distinguish between these boot stages, we use the retention properties of different memory regions:

- **FLASH** – Flash memory is completely overwritten when reflashed. If a magic value at a specific flash address is missing, we know the system was just flashed.
- **RESET** – Since RAM is retained during a soft reset, we store a magic number at a particular RAM address. If it's still there after a reboot, we've had a soft reset.
- **POWER** – At first power-on, the `POWER.RESETREAS` register is set to . This register tracks reset causes, so technically, we could use it to detect soft resets as well—but we prefer the magic number solution.
- **BOOT** – If none of the above conditions apply, we assume we're in a normal boot state.

All of this logic is encapsulated in the `boot` namespace, providing a simple way to determine the system's current state and react accordingly.

```
include/code/boot.h
```

```
1 namespace boot
2 {
3 boot_lev_t lev(); /* tells us the current boot level */
4 void init(); /* discovers what the current level is */
5 } // namespace boot
```

As a demonstration, here's what Apollo-RTOS prints on boot:

```
boot level: flash
boot stat: n_flash = 395, n_boot = 22, n_power = 19, n_reset = 114
```

This output shows the current boot level (flash) and the number of times the system has gone through different reset modes since the last FRAM file system format (one week ago).

7.2 Error handling

The scary part about not having a memory protection unit for the RAM is that a process can virtually rewrite any memory address. So in situations where we would really like to get a `segfault`, we will get silence, only for the system to crash horribly long time later.

So it's up to the programmer to ensure that the programs are free of bugs. The best way to ensure that is by having lots of checks throughout the code that checks parameters, results of calculations, and anything that could go wrong.

The C-header `<assert>` provides a macro `assert` that allows us to check that something holds, throwing an error if it doesn't. Since we don't have a mechanism for throwing errors in our microcontroller, we have to implement our own `assert`.

Before that, let's talk about how we debug things in a microcontroller. The only true answer to that question is: by printing stuff. So we need an implementation of `printf`. This `printf` implementation in Apollo-RTOS uses a buffer to send characters over the serial bus

asynchronously. So we also provide a blocking implementation in **kprintf**. We then can implement a convenience function **debug**, which only outputs stuff if the debug level is high enough.

```

                                include/utility/debug.h
1  template <debug_t level, typename... Args>
2  void debug(const char *fmt, Args... args)
3  {
4      if constexpr (level <= DEBUG_LEV) {
5          if constexpr (level <= ERROR)
6              kprintf(fmt, args...);
7          else
8              printf(fmt, args...);
9      }
10 }
```

With that, we can now implement a lightweight **assert** method. We use the macro as **assert**(expr, lev) expecting to cause a reset at the level lev if the expression expr evaluates to **false**. lev can be one of the three values:

- H_RESET: causes a hardware reset if the assertion is false,
- S_RESET: causes a soft reset,
- TERM: terminates the current process.

```

                                include/utility/debug.h
1  template <auto lev, typename... Args>
2  void __assert(bool ex, Args... args)
3  {
4      if (ex)
5          return; // everything under control
6
7      if constexpr (lev == H_RESET) {
8          if constexpr (sizeof...(args) != 0)
9              /* print debug information */
10
11          /* halt for manual intervention */
12      }
13
14      if constexpr (lev == S_RESET)
15          /* cause a soft reset */
16
17      if constexpr (lev == TERM)
18          /* terminate the current process */
19  }
20
21 #define assert(EX, LEV, ...) __assert<LEV>((EX), ##__VA_ARGS__)
```

7.3 Recovery

Now that we've handled the errors and caused an appropriate reset, we can talk about how the recovery mechanism actually works.

§8 Future Work

- (i) Potential improvements and extensions
- (ii) Unresolved challenges
- (iii) Research or application possibilities

§9 Conclusion

- (i) Summary of achievements
- (ii) Reflections on the AGC's enduring design principles
- (iii) Broader implications for OS design

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§A **Appendices**

- (i) Code snippets of key algorithms
- (ii) Detailed specifications
- (iii) Test results